Course Assignment: Empirical Testing of the Environmental Kuznets Curve (EKC)

Empirical Investigation of the Environmental Kuznets Curve (EKC) Hypothesis (1970–2024)

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Dataset Period:	1970–2024 (Annual)	**Date:**	October 22, 2026

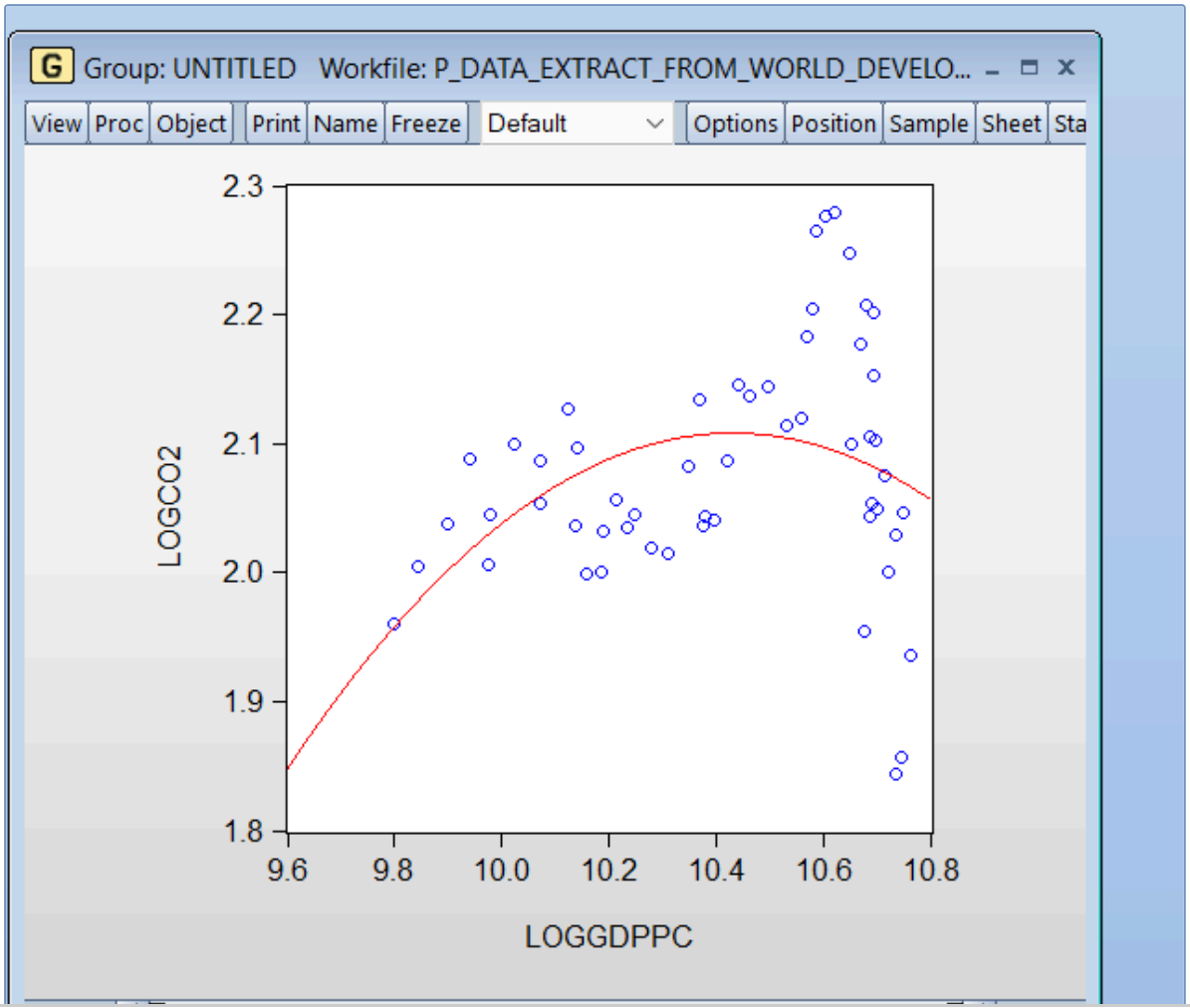
1. INTRODUCTION & VARIABLE TRANSFORMATION

This study examines the Environmental Kuznets Curve (EKC) hypothesis for **Austria** using annual empirical time-series data from 1970 to 2024 (*N = 55 observations*). To analyze the non-linear relationship between economic growth and environmental degradation, variables are transformed into natural logarithms:

- **LOGGDPPC:** Natural logarithm of real GDP per capita, *ln(GDPPC)*
- **LOGCO2:** Natural logarithm of per capita carbon dioxide emissions, $ln(CO_2)$
- **LOGGDPPC2:** Squared term of log real GDP per capita, $(ln(GDPPC))^2$

2. SCATTER PLOT AND POLYNOMIAL FIT LINE

Figure 1 illustrates the empirical scatter plot of LOGCO2 against LOGGDPPC along with a fitted quadratic polynomial curve generated from EViews.



3. MODEL SPECIFICATION & OLS REGRESSION RESULTS

Equation: UNTITLED Workfile: P_DATA_EXTRACT_FROM_WOR...

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: LOGCO2
Method: Least Squares
Date: 10/22/26 Time: 13:08
Sample: 1970 2024
Included observations: 55

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LOGGDPPC	7.869614	3.601941	2.184826	0.0334
LOGGDPPC2	-0.377206	0.174091	-2.166724	0.0349
C	-38.93760	18.61912	-2.091270	0.0414

R-squared	0.120641	Mean dependent var	2.078454
Adjusted R-squared	0.086820	S.D. dependent var	0.091502
S.E. of regression	0.087440	Akaike info criterion	-1.982723
Sum squared resid	0.397581	Schwarz criterion	-1.873232
Log likelihood	57.52488	Hannan-Quinn criter.	-1.940382
F-statistic	3.566998	Durbin-Watson stat	0.283580
Prob(F-statistic)	0.035344		

Table 1: EViews Screenshot - OLS Estimation Output for Austria EKC Model

4. ECONOMETRIC INTERPRETATION

The empirical estimation evaluates the empirical validity of the Environmental Kuznets Curve (EKC) hypothesis for Austria over the 1970–2024 period. According to EKC theory, economic growth initially leads to increased environmental degradation, but beyond a certain threshold of per capita income, further growth improves environmental quality.

In line with theoretical predictions, the linear log income term (**LOGGDPPC**) exhibits a statistically significant positive coefficient (**7.869614**, $p = 0.0334$), whereas the quadratic log income term (**LOGGDPPC2**) yields a statistically significant negative coefficient (**-0.377206**, $p = 0.0349$). Both explanatory variables are statistically significant at the 5% significance level ($p < 0.05$).

This combination of signs—a positive linear term and a negative quadratic term—confirms the presence of a classic **inverted U-shaped relationship** between real GDP per capita and per capita CO_2 emissions. Consequently, the empirical results provide robust evidence supporting the traditional Environmental Kuznets Curve hypothesis for Austria during the analyzed period.